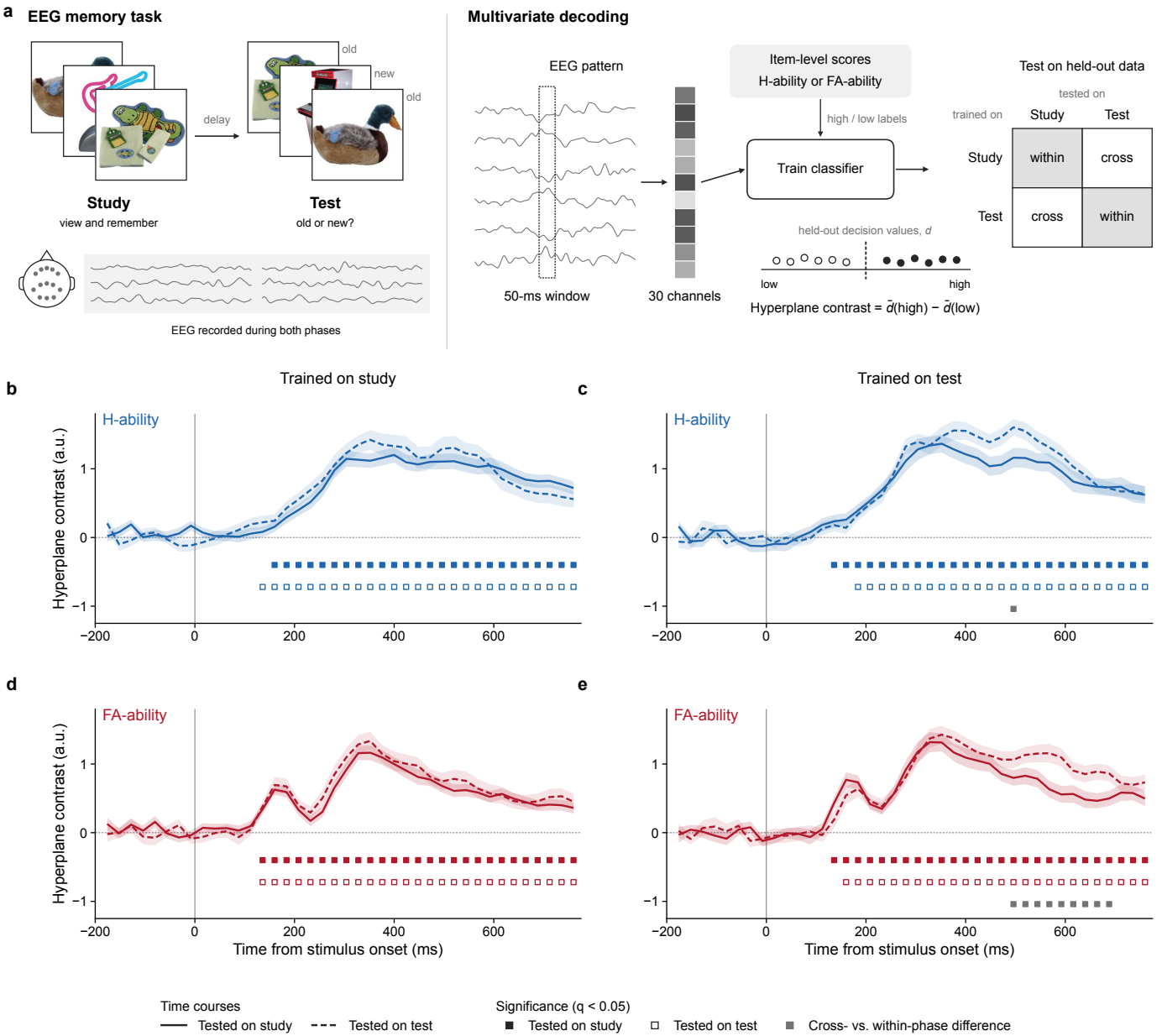

a
EEG memory task
old
new
old
delay
Study
view and remember
Test
old or new?
EEG recorded during both phases
Multivariate decoding
EEG pattern
50-ms window
30 channels
Item-level scores
H-ability or FA-ability
high / low labels
Train classifier
held-out decision values, d
low
high
Hyperplane contrast = d̄(high) − d̄(low)
Test on held-out data
tested on
trained on
Study
Test
Study
Test
within
cross
cross
within
b
Trained on study
H-ability
Hyperplane contrast (a.u.)
c
Trained on test
H-ability
d
FA-ability
Hyperplane contrast (a.u.)
Time from stimulus onset (ms)
e
FA-ability
Time from stimulus onset (ms)
Time courses
Tested on study
Tested on test
Significance (q < 0.05)
Tested on study
Tested on test
Cross- vs. within-phase difference